

## a Three layers that must not be conflated

Design blueprint: Top-Z graph + Z inputs for BOTH spots and genes

Public-code baseline: sampled edges + gene counts; A/C still use spatial batching

Input preflight completed; KL target, halo batching and reseeded remain declared conventions

| Protocol               | Expression-derived input             | Module comparison                            | Evidence scope                      |
|------------------------|--------------------------------------|----------------------------------------------|-------------------------------------|
| Blueprint              | Top-Z graph<br>Spot Z + gene Z       | A / B / C / D<br>Spatial and dynamic factors | Target design                       |
| Public-code baseline   | Sampled nonzero edges<br>Gene counts | A / B / C completed<br>D not run             | Three-seed module results           |
| Input-wiring preflight | Top-Z graph<br>Spot Z + gene Z       | A / B / C / D<br>Short runs only             | Wiring checked<br>Not effectiveness |

## b What has actually been run (3-seed results)

| Run | Protocol | Module       | rare-F1 | ARI    |
|-----|----------|--------------|---------|--------|
| A   | public   | fixed labels | 0.0000  | 0.3300 |
| B   | public   | + spatial    | 0.0000  | 0.3298 |
| C   | public   | + dynamic    | 0.0594  | 0.1334 |
| D   | not run  | not run      | --      | --     |

## c Current status

Strict-input effectiveness: NOT TESTED

Confidence mechanism: FAIL

Own-initial endpoint: FAIL

Public C: K0 = 13/9/11; D not run

C final - initial rare-F1:  $-0.0452 \pm 0.0708$ . C-A compares the full dynamic workflow, not assignment refresh alone.

Tau preflight: 8 candidates, 0.002 to 0.05, delta=0.8. No conclusion about ALL tau.